

PERSONAL DETAILS

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☎ +33 6 72 46 56 04
📍 Aix-en-Provence, France
🌐 lg-espace.github.io/portfolio/en.html
20 years old

IT SKILLS

Software

Microsoft Office (Word, Excel, PowerPoint), Canva, L^AT_EX, Git/GitHub, SolidWorks, CATIA, Granta

Programming languages

Python ●●●●●
C++ ●●●●●
MATLAB/Octave ●●●●●
SQL ●●●●●
Fortran ●●●●●
HTML/CSS ●●●●●
Self-assessed, from basic to advanced

LANGUAGES

French: mother tongue
English: upper-intermediate (B2), Cambridge English certificate
Spanish: intermediate (B1), DELE certificate

SOFT SKILLS

Teamwork
Project management
Time management
Giving presentations
Written communication
Rigour and analytical thinking

INTERESTS

Water polo: sports-study programme at school; team spirit and discipline.
Sailing and skiing
Mountain biking, golf, horse riding
Krav maga
Trombone

REFERENCES

Mr Cédric Galusinski
Head of the MOCA track
SeaTech, University of Toulon
✉ cedric.galusinski@univ-tln.fr
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Mr Stéphane Castells
Regional Manager
Géoterria
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Lysandre GOUDON

Engineering student (M.Eng.) in Computational Fluid Dynamics

PROFILE

Second-year engineering student at SeaTech, specialising in numerical modelling and fluid mechanics applied to the marine sector. Seeking an internship in **CFD, numerical simulation or ocean engineering**. Strong in mathematics, programming and data analysis.

EDUCATION AND QUALIFICATIONS

2025 – 2028 (expected) Toulon, France
SeaTech, School of Engineering, University of Toulon 

- Preparing an **M.Eng. in Computational Fluid Dynamics**: Modelling and Computation of Fluids and Structures
- **Subjects:** numerical analysis, partial differential equations (PDEs), fluid mechanics, wave physics, signal analysis, databases
- **Minor in Security and Defence:** geostrategy, infrastructure resilience, dual-use (maritime and defence) technologies

2023 – 2025 Lyon, France
Classes Préparatoires aux Grandes Écoles, Lycée Aux Lazaristes 

Intensive pre-engineering programme for entrance into elite engineering schools (PC track)

- **Subjects:** physics, mathematics, chemistry
- Carried out a supervised research project (TIPE) on supercapacitor energy density

2022 – 2023 Aix-en-Provence, France
ACESUP: early preparation for CPGE

- Intensive pre-course for scientific preparatory classes

2020 – 2023 Aix-en-Provence, France
Lycée Sainte-Catherine-de-Sienne, A-Levels equivalent

- French *Baccalauréat* (general), **with high honors**
- Subjects: mathematics, physics and chemistry; European section

WORK EXPERIENCE

Work placement

2019 – 2026
Project assistant: GAL Castagniccia Mare à Monti
Local Action Group (LAG) funded by the European LEADER programme

- Worked on the renovation of old mountain houses to create tourist accommodation
- Contributed to a rural development project supported by European funding

Internship

Geotechnical intern: Géoterria (geotechnical consultancy) 

- Studied and characterised soils; analysed building foundations
- Applied continuum mechanics principles to real sites

Projects

Mobile oceanographic probe: Astrolabe Expeditions 

- Designed a coastal-monitoring measurement system for a research association
- Developed a project website
- *Tools: CAD, Python, project management*

Numerical analysis projects
PDEs, ODEs and interpolation

- Implemented finite-difference schemes for the transport equation; analysed stability and accuracy
- Compared numerical methods: ODE solvers, Newton's algorithm, quadrature, Runge's phenomenon
- *Tools: Python, MATLAB/Octave*

Ocean physics projects

Waves and tides

- Studied wave reflection on a sloping plane; determined the Iribarren number experimentally
- Analysed tidal signals (HTM-NET network) using harmonic decomposition and FFT
- *Tools: Python, signal processing*

Data and materials projects

Databases, data science and materials

- Designed a relational database for the French railway network (SNCF) in SQL
- Analysed Energy Performance Certificate (DPE) data with Python
- *Tools: SQL, Python*